



PROJECT MAIEUTIC / 學問

From Inquiry to Software

The CityUHK Initiative on
Specification-Driven & Agentic Computing Education

“The art of bringing
latent knowledge
into light through
questioning.”



FROM FIRST YEAR ONWARD,
STUDENTS LEARN TO
**THINK, SPECIFY, AND
BUILD WITH AI AGENTS.**

SOCRATIC WATCHDOG

A coding tutor that asks, never tells.

It watches every cell you run in Jupyter. Wrong code gets **one spoken question**. Right code gets silence and confetti.

```
$ pip install socratic-watchdog
```

★ **FIRST-YEAR PROJECT**

Designed & shipped to PyPI by a
Year-1 Computer Science student

- ▶ Python · Jupyter · Colab · VS Code
- ▶ Any OpenAI-compatible LLM
- ▶ Speech: espeak-ng · Edge TTS · Kokoro
- ▶ Open source · MIT

01 THE PROBLEM

```
"My fib() crashes, fix it pls"
```

```
def fib(n):  
    if n < 2: return n  
    return fib(n-1) + fib(n-2)
```

The code runs.
The lesson doesn't.

02 WHAT THE STUDENT SEES

```
[3]: %%socratic  
def fib(n):  
    return fib(n-1) + fib(n-2)
```

RecursionError: maximum recursion depth exceeded

Socrates asks: "I see your function calls itself – what condition would stop it?"

```
[4]: %%socratic  
def fib(n):  
    if n < 2: return n # student's own fix  
    return fib(n-1) + fib(n-2)
```

Socrates says: Well done! You are thinking clearly.

→ spoken aloud + on screen → one question, never the fix → silent when correct

03 HOW IT WORKS

1 Run hidden tests first
pass → confetti, no AI call at all

2 Fail → ask the LLM as Socrates
code + error + failing tests

3 Speak one guiding question
sharper each retry, never stuck

0 answers given away

1 dependency (ipython)

3 voices, local or cloud

4 notebook platforms, incl. Colab & VS Code

Nightly report flags students stuck in a loop



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when built · now Year 2

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Code · GitHub



Try it · Colab

REFERENCES

[1] C. Zhao, Z. Wang, A. A. Ahmedani, B. Karimtayev, Marcello-Raphael-Christiano, N. M. B. Lua, R. Khamza, and C. Chan, "DIVE into agentic coding: Can you teach AI to teach you?" in Proc. IEEE Int. Conf. Teaching, Assessment, and Learning for Engineering (TALE), 2026. Accepted.

[2] R. Khamza, socratic-watchdog (v0.4.1), PyPI, 2026. pypi.org/project/socratic-watchdog

